

Alan Szmyt

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[LinkedIn](#) | [GitHub](#) | [ORCID](#)

Summary

Software engineer and systems architect building AI-native workflows, platform systems, and scalable software. Delivered mission-critical applications for government and research sponsors across backend, mobile, data, and infrastructure domains. Applies systems thinking and automation to turn complex workflows into reliable, maintainable products.

Experience

Software Engineer, Associate Staff

MIT Lincoln Laboratory

Lexington, MA

Apr 2019 – Aug 2025

- Led development of a mobile tracking system for USAID to monitor commodities in low-connectivity environments, successfully field-testing pilot shipments to Africa and enabling end-to-end supply chain visibility.
- Delivered a humanitarian deconfliction notification platform enabling analysts to visualize no-strike zones and submit secure, auditable reports, contributing to a follow-on effort that secured approximately \$1M in additional funding.
- Built a counter-human trafficking investigation tool that enabled law enforcement analysts to triage large volumes of digital evidence and reduce time to actionable leads.
- Developed externally accessible geospatial decision-support applications integrating shared data hubs, interactive mapping, and domain-specific datasets to improve situational awareness and analytical decision-making.
- Contributed to NATO disaster-response exercises by extending field-ready mobile capabilities, including hazard visualization, area-of-interest management, and real-time location reporting across multinational operations.
- Designed reproducible development environments and optimized Android (Gradle, Java/Kotlin) and modern web (Vite/ESBuild) build pipelines, improving onboarding speed, build reproducibility, and developer feedback cycles.
- Developed full-stack geospatial systems for government and research sponsors, integrating data engineering, visualization, and automation for mission-critical analysis and field operations.

Founder & Systems Architect

Incompris LLC

Wilmington, MA

2023 – Present

- Founded and operate an independent creative-technology studio focused on AI-assisted workflow systems, deterministic publishing pipelines, and multimodal content automation.
- Architected a specification-driven publishing platform that transforms canonical creative assets into reproducible multi-format artifacts (PDF, CBZ, WebP, AVIF, and LaTeX) through staged workflows and validation pipelines.
- Designed AI-native workflow architectures that treat language models as bounded system components with explicit contracts, synchronization checkpoints, and reproducible execution paths.
- Built multimodal content-generation pipelines integrating LLMs, image generation, video automation, metadata processing, and publication tooling into cohesive end-to-end workflows.
- Developed local-first development environments and CI-backed automation systems using Docker, GitHub Actions, and reproducible infrastructure patterns to accelerate development and artifact validation.
- Authored reflector, an independent research publication exploring synchronization architectures, recursive AI-assisted software engineering workflows, and human-AI governance systems.

Publications

- **reflector: Reflective Synchronization Systems for Recursive AI-Assisted Software Engineering** (2026). Independent research publication proposing a synchronization architecture for AI-assisted software engineering workflows, including recursive drift, reflective auditing, synchronization checkpoints, and specification-driven orchestration. Published through Zenodo with DOI: [10.5281/zenodo.20477044](https://doi.org/10.5281/zenodo.20477044).
Publication: <https://zenodo.org/records/20477045>
ORCID: <https://orcid.org/0009-0008-5291-9795>

Education

M.S., Software Development
Boston University

Boston, MA
2023

- Focused on software engineering, full-stack development, and system architecture.
- Capstone projects included a WAV metadata parser in Python, an Android beatmaker application using JNI and C++, and a personalized music recommendation system incorporating algorithmic ranking techniques.

B.S., Computer Science (Minor in Mathematics)
University of Massachusetts Lowell

Lowell, MA
2017

- Academic focus included Operating Systems, Data Structures, Algorithms, Computer Graphics, and Discrete Mathematics.
- Participated in independent research exploring decentralized blockchain networks.

Skills

Languages: Python, TypeScript, Java, Kotlin, Dart, C++, SQL, Bash

Frontend & Mobile: React, Redux Toolkit, RTK Query, Flutter, Android, JavaScript, HTML/CSS

Backend & Data: Flask, PostgreSQL, MySQL, Elasticsearch, GeoServer, REST APIs, GraphQL, Distributed Systems

Platform & Infrastructure: Docker, Kubernetes, GitHub Actions, Devcontainers, Pulumi, Linux, CI/CD

Geospatial & Visualization: deck.gl, Kepler.gl, GeoJSON, Mapbox, Data Visualization

AI & Automation: LLM Workflows, AI-Assisted Development, Workflow Automation, Specification-Driven Systems, Publication Pipelines